Scenarios, Goal Modeling and Refinement

17-626 Requirements for Information Systems

Prof. Travis Breaux

What are goals?

- Goals are prescriptive statements about the intent of the system
  - Patient medical histories shall remain confidential
  - Each patient shall be notified about upcoming appointments
  - Pulsometers shall be medically accurate

- Goals are satisfied through cooperating agents
  - Agents include people and software, such as:
    - Patient
    - Medical record
    - Pulsometers, Fitness Monitors, …

Domain properties and assumptions

- **Domain properties** are descriptions of the problem world that are expected to hold, invariably
  - A medical appointment consists of one patient and one doctor

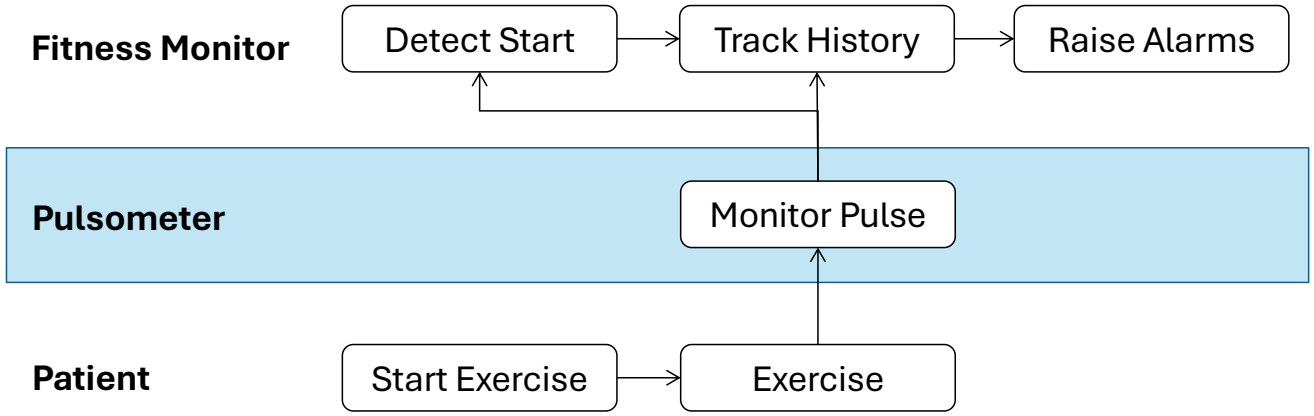
There are two types of assumptions…

- **Hypotheses** are descriptions satisfied by the environment and subject to change
  - Each patient has at most one primary care physician
- **Expectations** are prescriptions assigned to the responsibility of a single agent in the environment
  - Patients shall carry and present their Medical Insurance Card

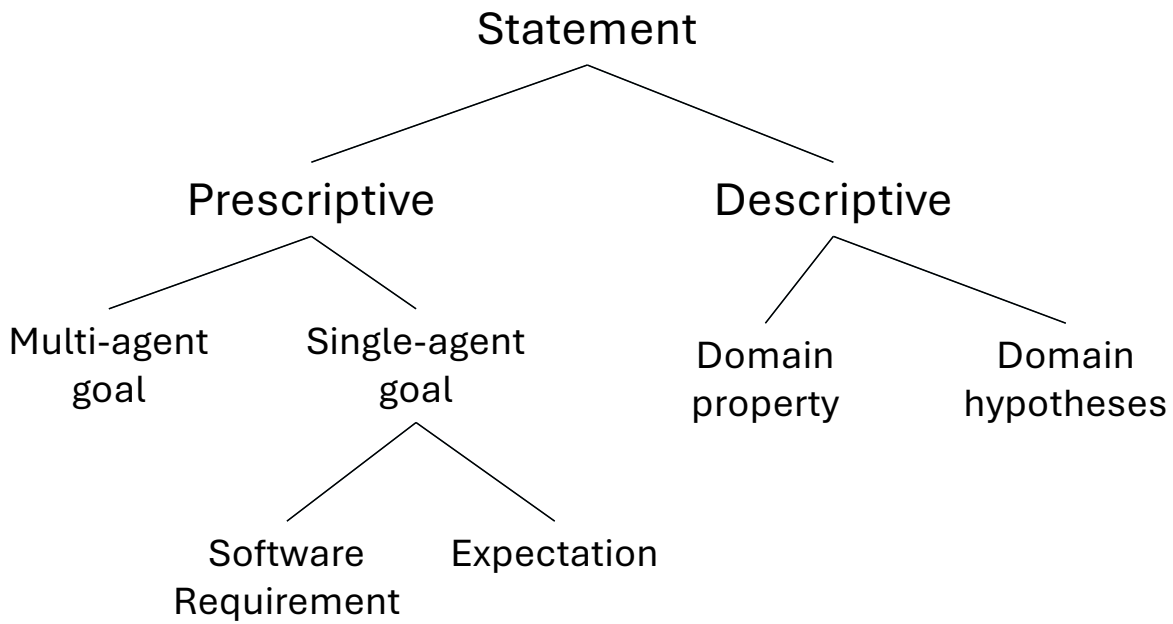
- *How do expectations differ from system requirements?*

System v. agent behaviors

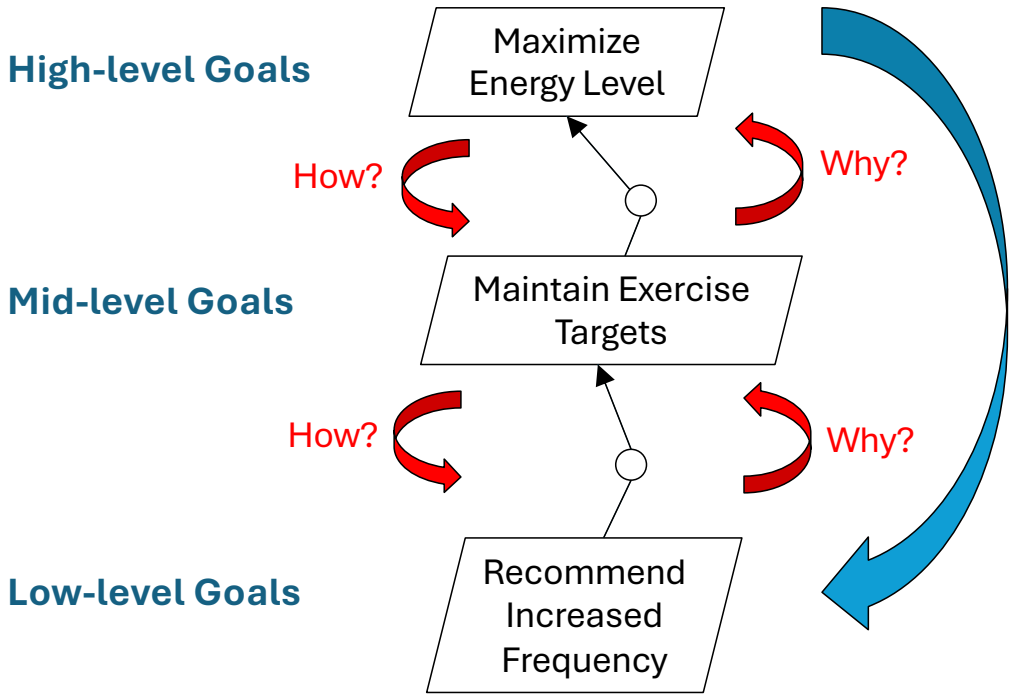
- *System behaviors* describe parallel behaviors of agents that form the system
- *Agent behavior* is captured by a sequence of state transitions for the items that the agent controls or observes



Putting it all together



High-level to low-level goals



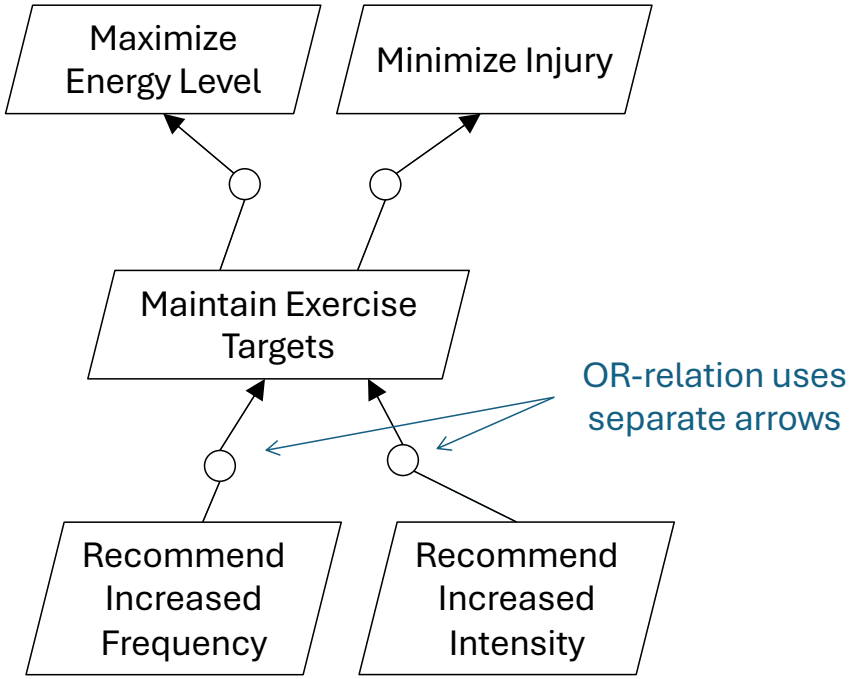
- High-level goals constrain low-level goals
- High-level goals provide rational for low-level goals

High-level to low-level goals

High-level Goals

Mid-level Goals

Low-level Goals

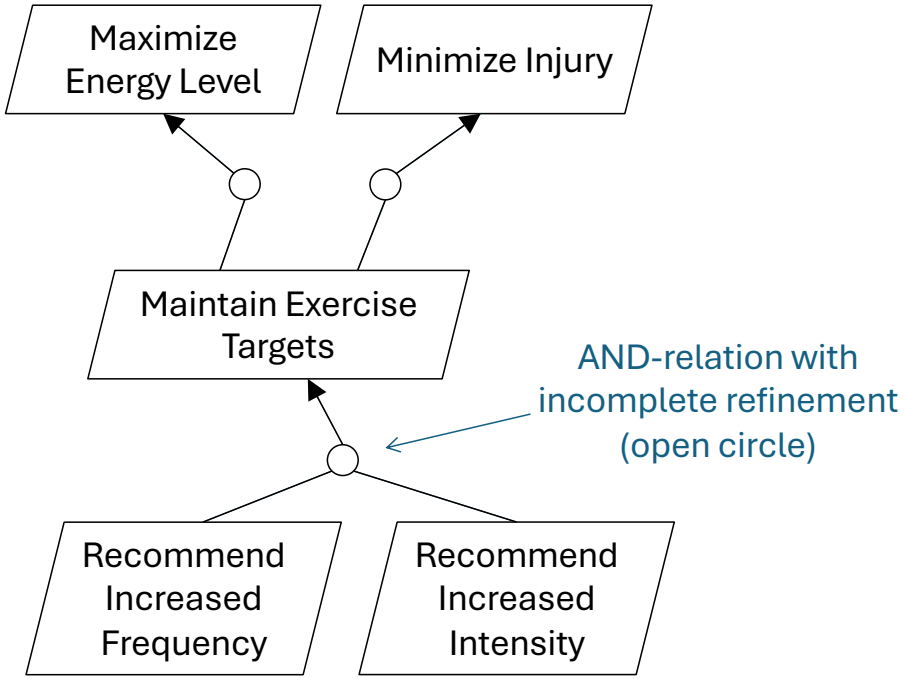


High-level to low-level goals

High-level Goals

Mid-level Goals

Low-level Goals

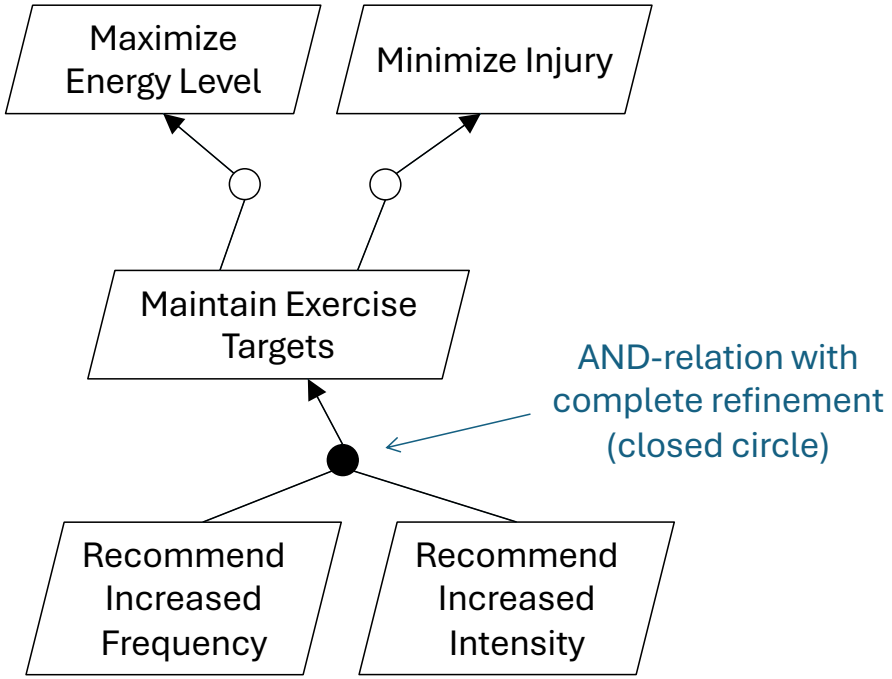


High-level to low-level goals

High-level Goals

Mid-level Goals

Low-level Goals



Goal AND-refinement

- **Complete**: The satisfaction of all sub-goals is sufficient for the satisfaction of the parent goal
- **Consistent**: The sub-goals, domain properties and hypotheses may not contradict each other
- **Minimal**: If one of the sub-goals is missing, the satisfaction of the parent goal is no longer guaranteed

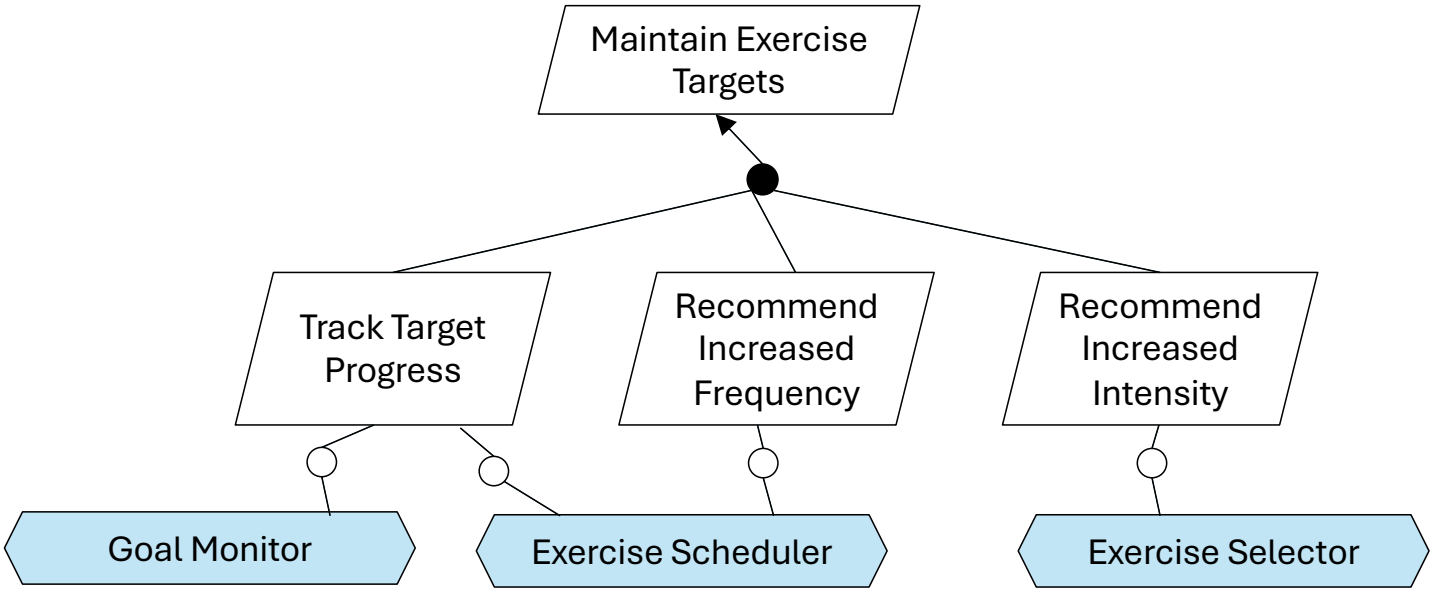
Goal OR-refinement

- Assignment to different agents has varying effects
- For example: to which agent do we assign the goal Raise Alarms?
  - **Fitness Monitor**
  - **Exercise Specialist**

Goal OR-refinement

- Assignment to different agents has varying effects
- For example: to which agent do we assign the goal Raise Alarms?
  - **Fitness Monitor** – may reduce response times, but increase development costs and produce false alarms, or missed alarms
  - **Exercise Specialist** – may detect complex anomalies, but increase latency and create an increase load for this agent, thus delaying other tasks

Goal and agent assignment

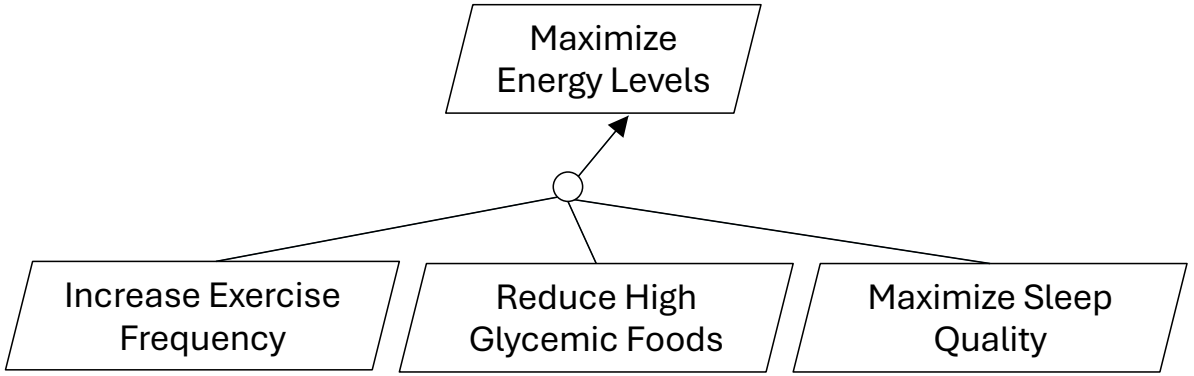


Behavioral Goals

- **Achievement** goals prescribe an intended behavior where a target condition must sooner or later hold
  - Achieve [ExerciseFrequencyIncreased]
  - Cease [Exercise] **if** [HeartRateMaximumExceeded]
- **Maintenance** goals prescribe where a “good” condition must always hold in every future state
  - Maintain [ExerciseTargetsReached]
- **Avoidance** goals prescribe where a “bad” condition must never hold in any future state
  - Avoid [Dehydration]

Soft goals

- Soft goals prescribe preferences among alternative system behaviors that are more or less fulfilled
- Satisficing describes when a goal is more or less satisfied by one alternative than another



Probabilistic goals

- Probabilistic goals are behavioral (not soft) goals that must be satisfied in $n\%$ of cases

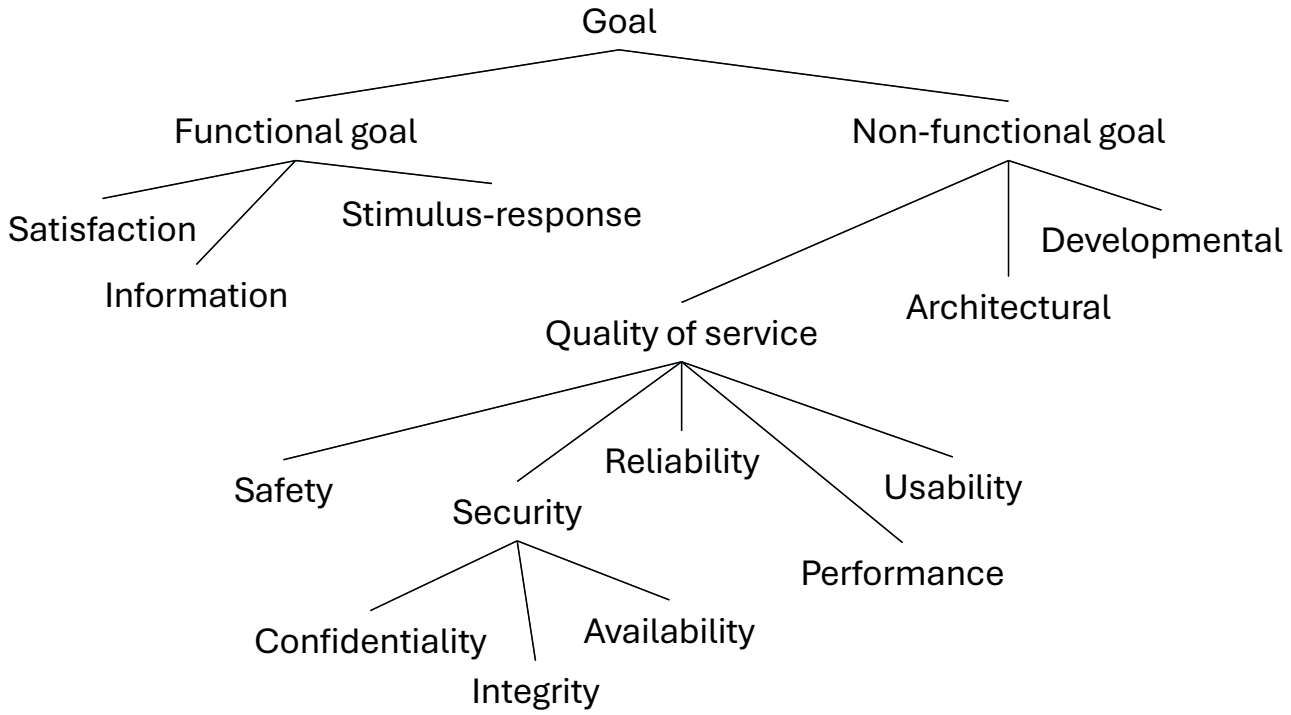
- Achieve [50% Target Heart Zone]
  - i.e., 50% of minutes spent exercising are within the target heart rate zone for the user based on their age

Identify the statement type

1. At least half of users will increase their exercise frequency within 30 days
2. Remote updates should not occur while a user is exercising
3. Weight is measured in pounds or kilograms
4. Heart rate must be logged every 10-15 seconds during exercise
5. Older users (age 65+) will complete a tutorial about safe exercise

Choices: Domain property, Domain hypothesis, Expectation, Avoidance goal, Achievement goal, Maintenance goal, Probabilistic goal

Goal categories



Identify goals in scenarios

Scenarios illustrate a “thread of execution” through the existing or envisioned system

Example Scenario:

“Today is Monday, and Patrick is ready to set a new health goal. He begins by planning his weekly meals, thinking about what he learned online about foods that boost his energy level. After creating his meal plan, he shares it with his dietician who reviews his workout routine from last week, plus the new goal that Patrick set for this week: an additional 30 minutes of walking, which the dietician translates into calories burned. The dietician proposes changes to the meal plan to account for the additional exercise, and Patrick reviews the changes.”



Patrick is an information seeker and believes in frequent exercise but is concerned about exceeding his limits. He lacks confidence about intense exercise.

Identify goals in scenarios

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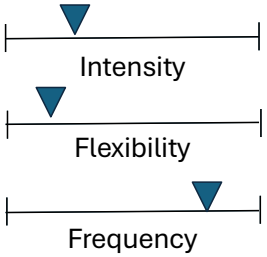
Candidate Goals:

- Set a new health goal
- Plan weekly meals
- Share meal plan with dietician
- Review workout routine
- Compute calories burned
- Propose meal plan changes
- Review proposed changes

Personas as stakeholder proxies



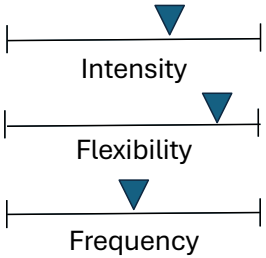
Name: Patrick
Age: 72
Gender: Male
Job: Retired Financial Analyst



Patrick is an information seeker, and believes in frequent exercise but is concerned about exceeding his limits. He lacks confidence about intense exercise.



Name: Allison
Age: 59
Gender: Female
Job: Aerospace Engineer



Allison changes her behavior to meet health goals, focuses on stress reduction, and is interested in modern diet options based on sound science.

Discovering goals from personas



Patrick is an information seeker, and believes in frequent exercise but is concerned about exceeding his limits. He lacks confidence about intense exercise.

- Read articles about diet and exercise, and link what he learns to changes in his meal and exercise plan [Plan using reputable sources]
- Compare his weekly performance against his current goals [Improve confidence]
- Compare his goals against his doctor's recommendations [Avoid exceeding limits]

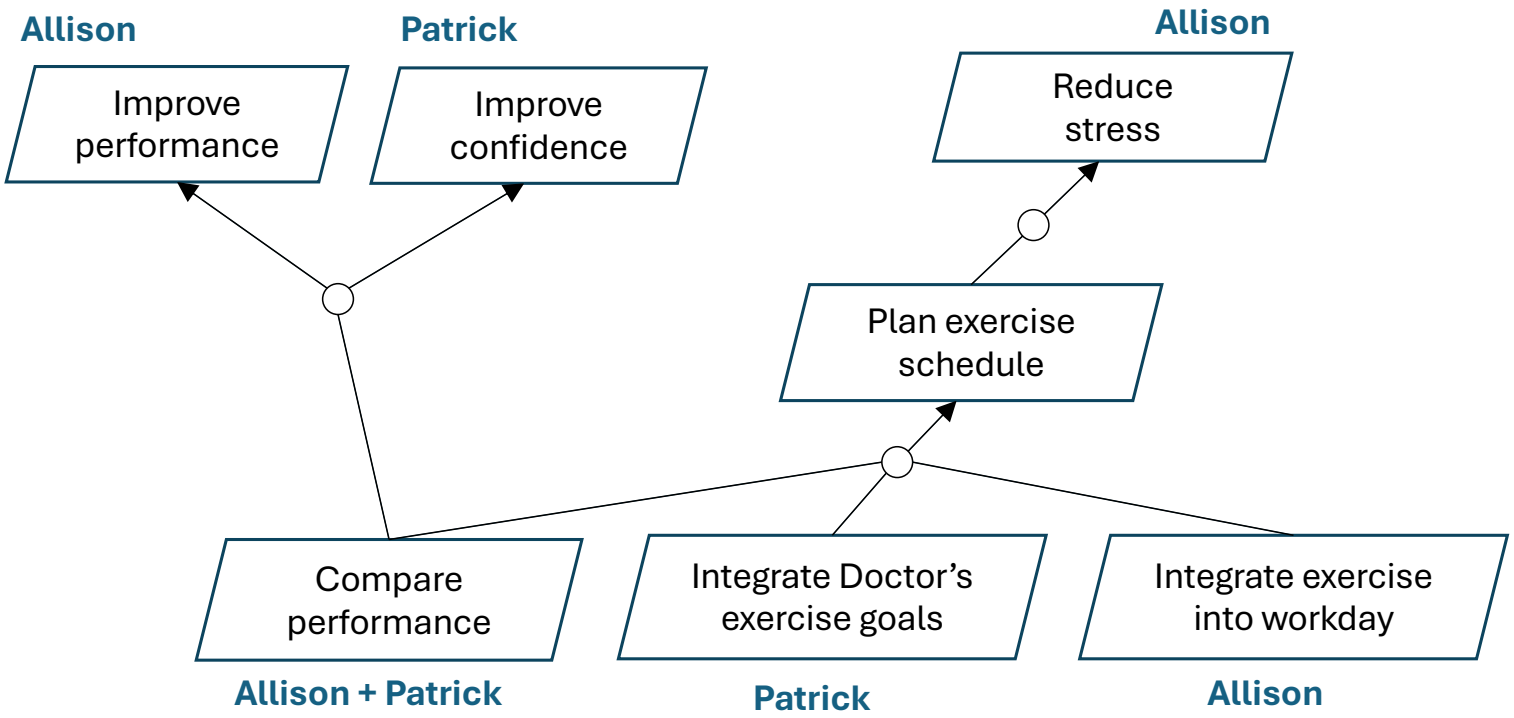
Discovering goals from personas



Allison changes her behavior to meet health goals, focuses on stress reduction, and is interested in modern diet options based on sound science.

- Compare her weekly performance against her current goals [Improve performance]
- Schedule exercise activities to breakup workday [Reduce stress]
- Read articles about diet, and link what she learns to changes in her meal plan [Plan using reputable sources]

Integrating goals from personas



Refinement Heuristic Summary

Heuristics for identifying goals…

- Identify strategic or business goals
- Deriving goals from elicitation materials
- Instantiate goal categories
- How / why goal exploration
- Identify agent’s wishes
- Splitting responsibilities
- Abstract goals to reach system boundary

Why conflicts arise?

- Different agents formulate goals
  - Increase exercise frequency (Exercise Specialist)
  - Reduce boredom during exercise (Patients)
- Goal boundaries are vague
  - Track exercise history (when does exercise start and end?)
  - Minimize injury (what physical effects constitute an injury?)
- Two soft goals prescribe an increase and decrease in the same quantity

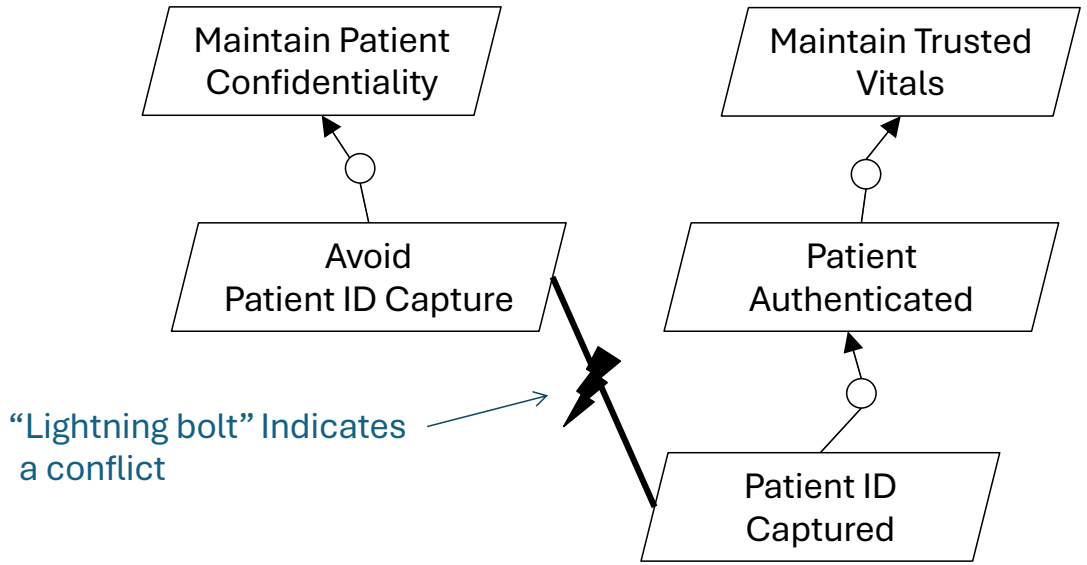
Types of conflict

- **Strong conflicts** involve goals that cannot be satisfied simultaneously
  - Patient information cannot be disclosed outside medical record
  - Exercise specialist must know history of patient vitals

- **Weak conflicts** occur in boundary cases
  - Patient should exercise four times per week
  - Patient should accumulate 150 minutes of moderate exercise per week

Goal conflicts

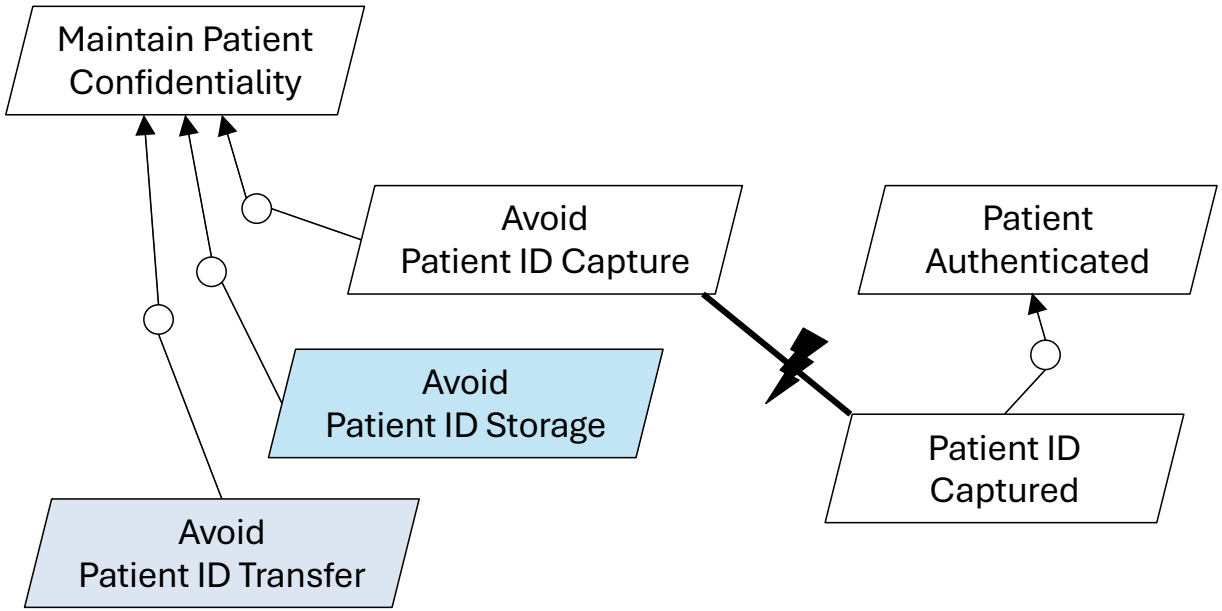
- Conflicts arise between goals at different levels in the goal hierarchy



Informal conflict detection

1. Maximize user engagement
2. Notify users of upcoming group exercises based on their location
3. Minimize user dissatisfaction
4. Cache user password for faster recording of vitals in medical record
5. Allow users to be anonymous
6. Increase exercise intensity
7. Allow users to post messages about their exercises

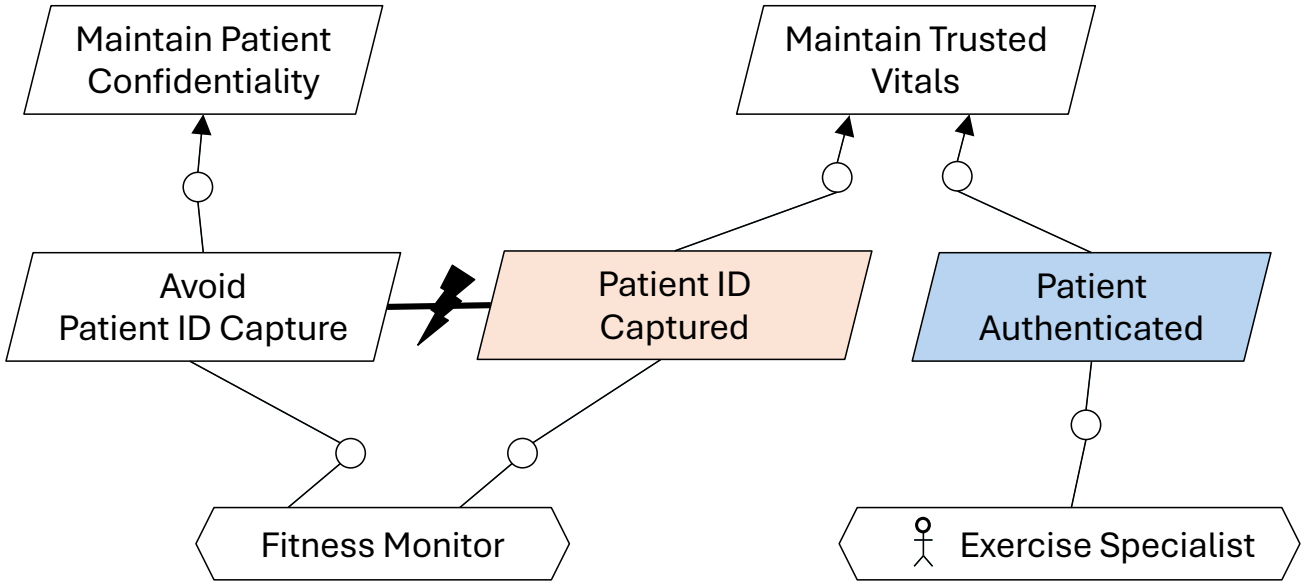
Weakening goals to resolve conflicts



Strengthening goals for risks

- Unacceptable exposure to risks may require strengthening goals
- Ensuring patient confidentiality involves different risk levels
  - Avoid[Patient ID Capture] – minimal risk, because only vitals are recorded, and not the ID
  - Avoid[Patient ID Storage] – higher risk, because one-time substitute ID is generated
  - Avoid[Patient ID Transfer] – highest risk, because patient ID is stored on user device

Transfer the conflict outside the system

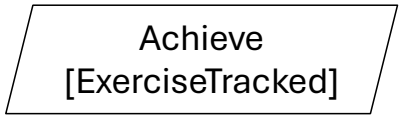


Conflict v. Trade-off

- What's the difference between a conflict and a trade-off?

Refinement Patterns

Milestone-driven pattern

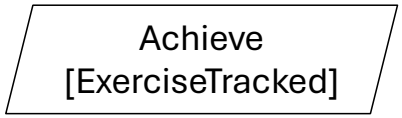


- Behavioral achievement goals may include “intermediate” conditions, which identify as necessary milestones to achieving the target condition

Achieve [TargetCondition]

if [**CurrentCondition then**] sooner-or-later **TargetCondition**

Milestone-driven pattern



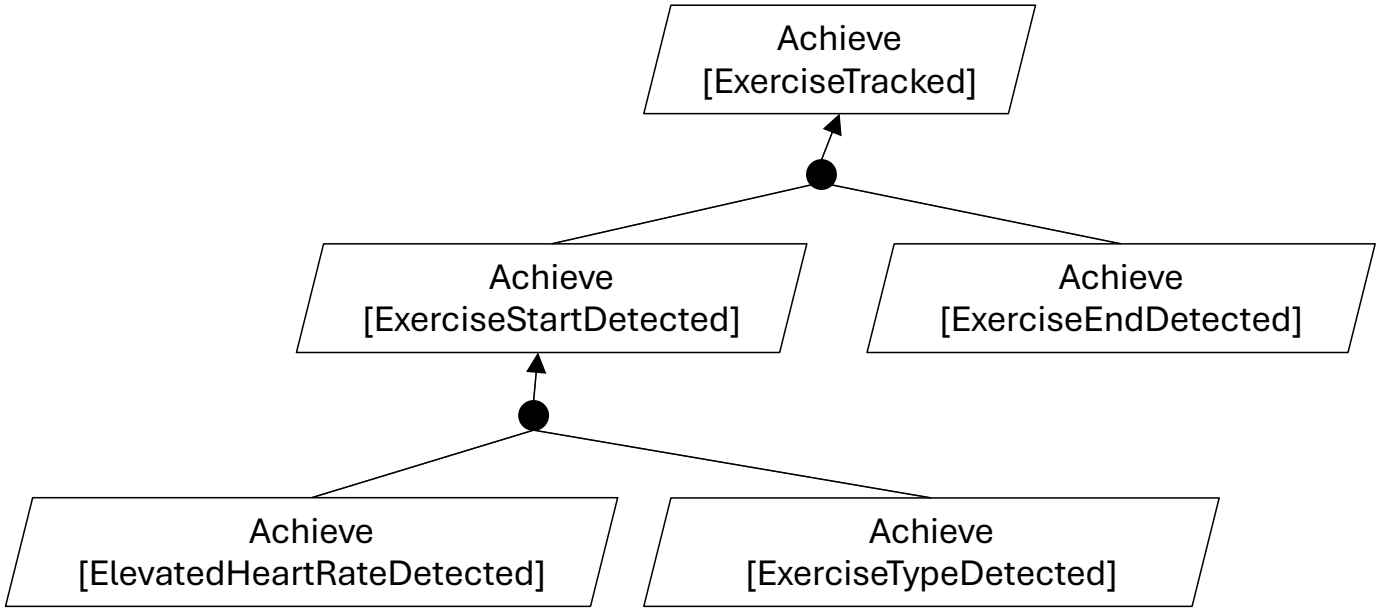
- Behavioral achievement goals may include “intermediate” conditions, which identify as necessary milestones to achieving the target condition

Achieve [ExerciseTracked]

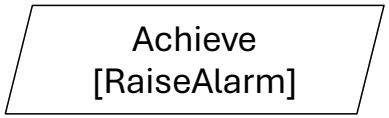
if [**HeartRateElevated then**] sooner-or-later **ExerciseTracked**

Is elevated heart rate necessary and sufficient, or is there something else we want to know?

Milestone-driven pattern



Decomposition-by-cases pattern



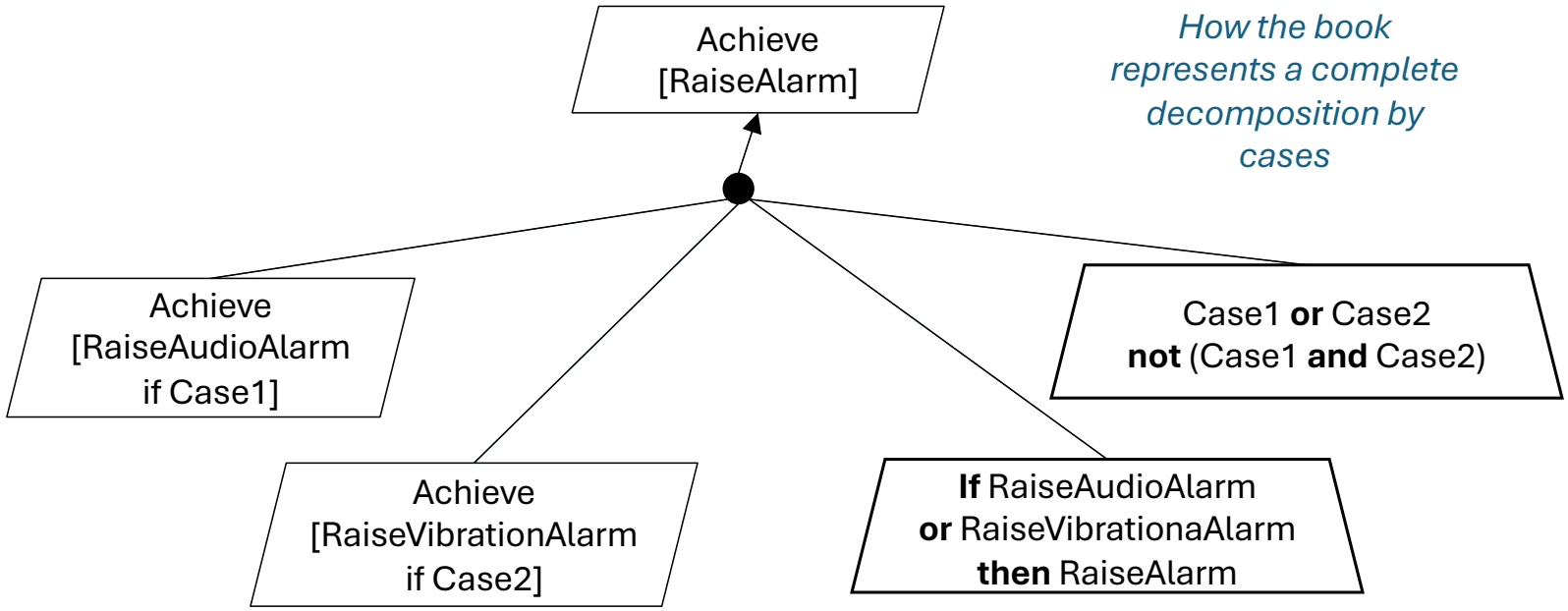
- Achievement goals that can be reached through different target conditions

Achieve [TargetCondition]

if **TargetCondition1** or **TargetCondition2** then **TargetCondition**

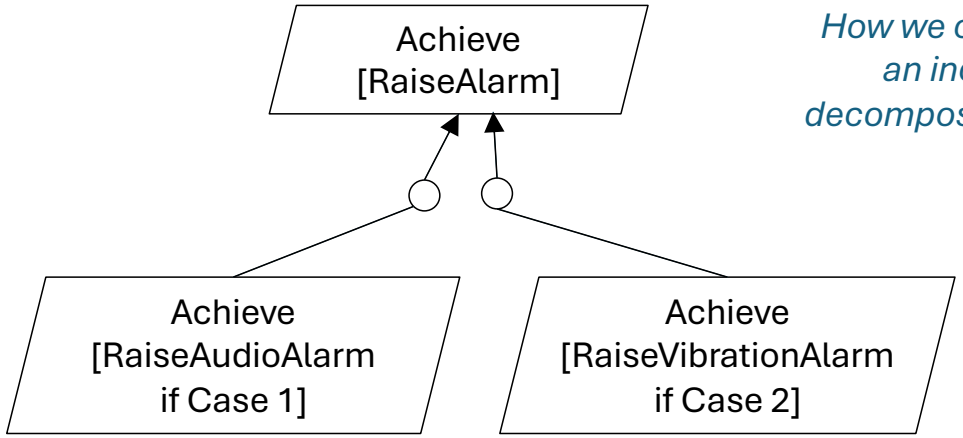
Case1 or **Case2 (**not **Case1 and Case2)**

Decomposition-by-cases



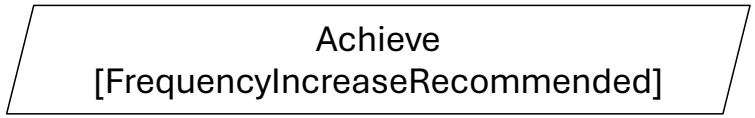
How the book represents a complete decomposition by cases

Decomposition-by-cases



How we can represent an incomplete decomposition by cases

Guard-introduction pattern



- Achievement goal wherein a guard condition must be set to reach target conditions
- **Guard condition** must be true before a system can transition to the next state

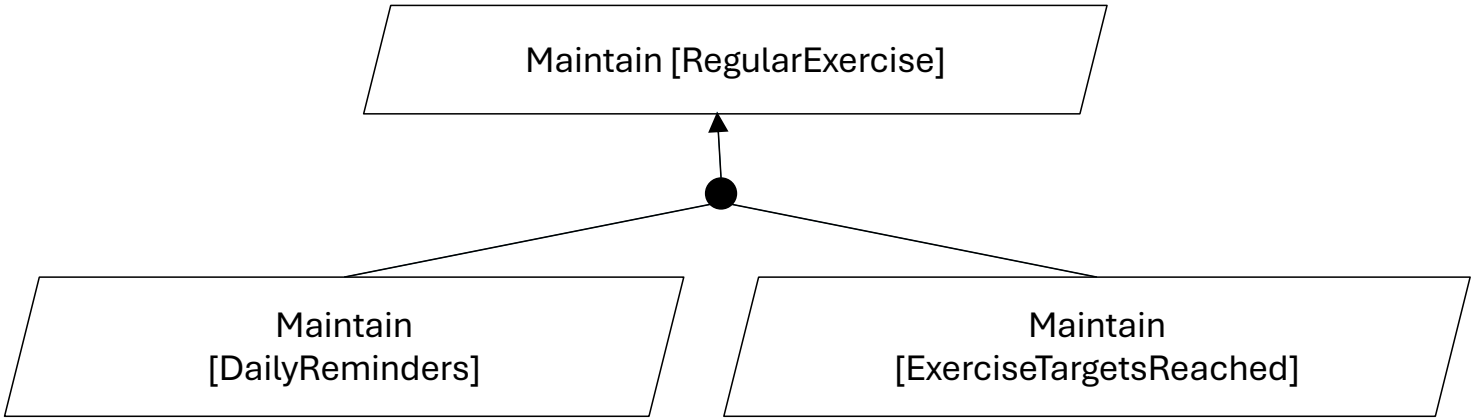
Achieve [FrequencyIncreaseRecommended
if ExerciseTargetsReached **and not** InjuryReported]

Divide-and-conquer pattern

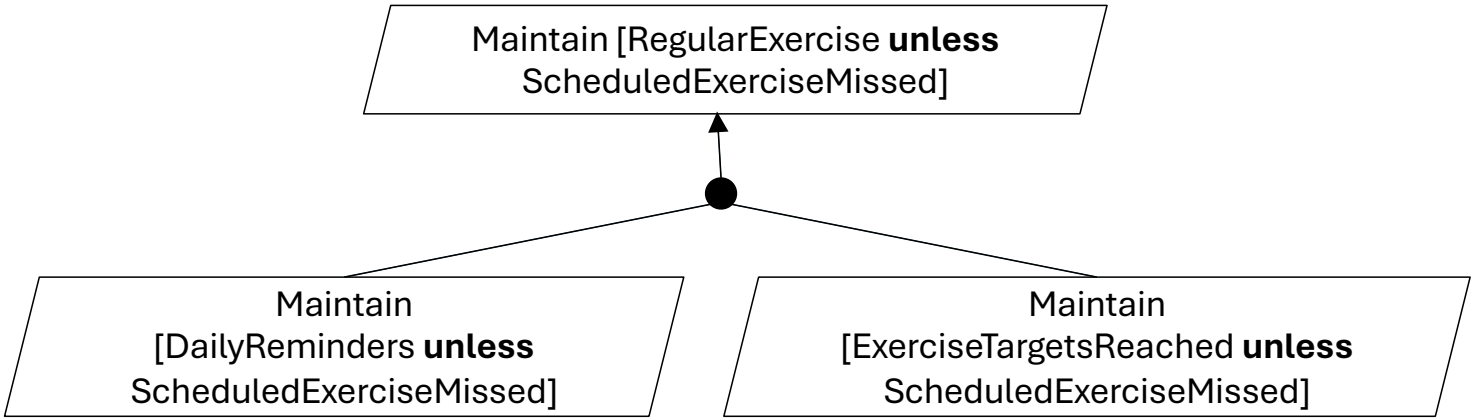


- Maintenance goal wherein a good condition is comprised of maintaining two conjoined conditions

Divide-and-conquer pattern



Divide-and-conquer pattern



Unrealizability Patterns

- **Unmonitorability pattern** – when goal condition cannot be ***monitored*** by responsible agent
- **Uncontrollability pattern** – when goal condition cannot be ***controlled*** by responsible agent

Lesson Summary

- What are goals?
- How goals map to system behaviors?
- Goal refinement
- Goal assignment to agents
- Goal types and categories
- Goal refinement heuristics